Chapter 4. Sampling Distributions

Binomial, normal, chi-square, t and F distributions

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4.1 The Bernoulli and normal distributions as population distributions

Motivation. Statistical inference judges an estimate by the probability distribution of the estimator, and that distribution is driven by the population distribution. The two most important population models are the Bernoulli distribution (a two-valued characteristic) and the normal distribution (a continuous characteristic).

4.1.1 Bernoulli trials

Many everyday experiments have exactly two outcomes: pass/fail, win/lose, cured/not cured. Such an experiment is called a **Bernoulli trial**; its outcomes are labelled success s and failure f.

Definition 4-1 (Bernoulli trial)

(a) A Bernoulli trial has exactly two outcomes, success s and failure f, so its sample space is the two-point set $S = \{s, f\}$.
(b) The success probability is $p = P(\{s\})$ and the failure probability is $q = P(\{f\}) = 1 - p$, where $0 \le p \le 1$.

Definition 4-2 (Bernoulli random variable)

On the sample space $S = \{s, f\}$ of a Bernoulli trial, the random variable X with $X(s) = 1$, $X(f) = 0$ is called a **Bernoulli random variable**. Its distribution, $P(X = 1) = p$, $P(X = 0) = 1 - p$, is the **Bernoulli distribution** with success probability p.

The Bernoulli distribution models a population whose characteristic is dichotomous (e.g. patients "cured"/"not cured" coded 1/0).

Example 4-1 (Mean and variance of the Bernoulli distribution)

Find the mean and variance of the Bernoulli distribution.
Solution. $E(X) = 1 \cdot p + 0 \cdot (1 - p) = p$ and $E(X^2) = 1^2 \cdot p + 0^2 \cdot (1 - p) = p$. Hence

$$\mathrm{Var}(X) = E(X^2) - \{E(X)\}^2 = p - p^2 = p(1 - p).$$

Bernoulli trials are **independent** if the result of one trial does not affect the others, and **repeated** if the success probability is the same in every trial.

Illustration 4-1: sampling with replacement

A deck of 10 cards contains 3 winning cards; two cards are drawn by simple random sampling *with* replacement. Let X_1, X_2 be the Bernoulli variables of the first and second draws (1 = winning card). Then

$$P(X_2 = 1) = P(X_2 = 1 \mid X_1 = 1) = P(X_2 = 1 \mid X_1 = 0) = \tfrac{3}{10},$$

and similarly $P(X_2 = 0) = \tfrac{7}{10}$ for either value of X_1. Also $P(X_1 = 1) = P(X_2 = 1) = \tfrac{3}{10}$, so the two trials are independent and repeated: X_1, X_2 are independent with the same distribution. More generally, n draws with replacement give independent, identically distributed $X_1, \ldots, X_n$.

Illustration 4-2: sampling without replacement

Draw the two cards of Illustration 4-1 *without* replacement. Then $P(X_1 = 1) = \frac{3}{10}$, $P(X_2 = 1 \mid X_1 = 1) = \frac{2}{9}$, $P(X_2 = 1 \mid X_1 = 0) = \frac{3}{9}$, and by the law of total probability

$$P(X_2 = 1) = \tfrac{2}{9} \cdot \tfrac{3}{10} + \tfrac{3}{9} \cdot \tfrac{7}{10} = \tfrac{3}{10}.$$

So each draw wins with probability $\frac{3}{10}$, but $P(X_2 = 1) \neq P(X_2 = 1 \mid X_1 = 1)$: the trials are not independent.

Illustration 4-3: general finite deck

With N cards of which M are winning, sampling two without replacement gives

$$P(X_1 = 1) = P(X_2 = 1) = \frac{M}{N}, \qquad P(X_2 = 1 \mid X_1 = 1) = \frac{M-1}{N-1},$$

so again the draws are identically distributed but dependent. If N (and M) is large relative to the sample, $P(X_2 = 1) \approx P(X_2 = 1 \mid X_1 = 1)$ and the draws are nearly independent.

Sampling with vs. without replacement

From a finite population, sampling without replacement is practically indistinguishable from sampling with replacement when the population size N is large compared with the sample size.

4.1.2 The normal distribution

The normal distribution is the leading model for an infinite population whose characteristic is continuous. Its density curve

(a) is symmetric about a single point, the population mean μ;

(b) is highest at the centre μ;

(c) decreases as one moves away from μ;

(d) has inflection points at $\mu \pm \sigma$.

Definition 4-3 (Normal density with mean μ and standard deviation σ)

$$p(x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}, \qquad -\infty < x < \infty.$$

We write $X \sim N(\mu, \sigma^2)$.

Changing μ shifts the curve without changing its shape; a smaller σ makes the curve narrower and taller near μ, a larger σ flatter. If one characteristic is observed at random from a normal population, the observation X has the population distribution, so $X \sim N(\mu, \sigma^2)$.

Theorem 4-1 (Properties of the normal distribution 1)

If $X \sim N(\mu, \sigma^2)$, then for any constants $a \neq 0$ and b,

$$aX + b \sim N(a\mu + b,\ a^2\sigma^2).$$

Proof. [Supplement] We use moment generating functions (mgf), $M_X(t) = E(e^{tX})$; an mgf finite near 0 determines the distribution.

Step 1. For $X \sim N(\mu, \sigma^2)$, substituting $x = \mu + \sigma z$ and completing the square $tz\sigma - \frac{z^2}{2} = -\frac{1}{2}(z - \sigma t)^2 + \frac{\sigma^2 t^2}{2}$,

$$M_X(t) = e^{\mu t} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{\sigma t z - z^2/2}\, dz = e^{\mu t + \sigma^2 t^2/2} \int_{-\infty}^{\infty} \frac{e^{-(z-\sigma t)^2/2}}{\sqrt{2\pi}}\, dz = e^{\mu t + \sigma^2 t^2/2}.$$

Step 2. Hence $M_{aX+b}(t) = e^{bt} M_X(at) = e^{(a\mu+b)t + a^2\sigma^2 t^2/2}$, which is the mgf of $N(a\mu + b, a^2\sigma^2)$. By uniqueness of mgfs the claim follows. ∎

(If $a = 0$ then $aX + b = b$ is constant, a "normal distribution with variance 0".)

In particular, taking $a = 1/\sigma$, $b = -\mu/\sigma$,

$$X \sim N(\mu, \sigma^2) \implies Z = \frac{X - \mu}{\sigma} \sim N(0, 1). \tag{4.1}$$

$N(0, 1)$ is the **standard normal distribution**; a standard normal variable is usually written Z. Appendix C tabulates $P(Z \le z)$.

Definition (Probabilities from the standard normal table)

(a) $P(a \le Z \le b) = P(Z \le b) - P(Z \le a)$.
(b) $P(Z \ge z) = 1 - P(Z < z)$.

Example 4-2 (Two tails)

Find $P(Z < -1.9 \text{ or } Z > 2.1)$.
Solution. The events are disjoint, so the probability is $P(Z < -1.9) + P(Z > 2.1) = 0.0287 + (1 - 0.9821) = 0.0287 + 0.0179 = 0.0466$.

Example 4-3 (A quantile)

Find z with $P(Z > z) = 0.025$.
Solution. $P(Z \le z) = 1 - 0.025 = 0.975$, so the table gives $z = 1.96$.

Definition (Upper quantile z_α)

For $0 < \alpha < 1$, z_α is the number with $P(Z > z_\alpha) = \alpha$, $Z \sim N(0, 1)$; it is the $100(1 - \alpha)$th percentile (quantile) of $N(0, 1)$. Frequently used values:

$$z_{0.005} = 2.58, \qquad z_{0.025} = 1.96, \qquad z_{0.05} = 1.645.$$

If $X \sim N(\mu, \sigma^2)$, standardizing with (4.1) gives

$$P(X \le b) = P\Big(Z \le \frac{b - \mu}{\sigma}\Big),$$
$$P(a \le X \le b) = P\Big(\frac{a - \mu}{\sigma} \le Z \le \frac{b - \mu}{\sigma}\Big).$$

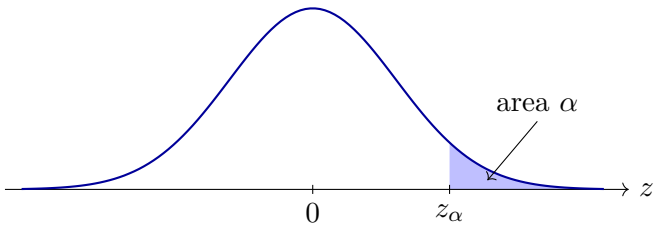


Figure 1: The upper quantile z_α of $N(0,1)$.

Example 4-4 (Proportion of A grades)

Grades in a statistics class are approximately $N(60, 10^2)$. Students scoring 75 or more get an A. Approximate the proportion of A grades.
Solution. $P(X \geq 75) = P(\frac{X-60}{10} \geq \frac{75-60}{10}) \approx P(Z \geq 1.5) = 0.0668$.

Theorem 4-2 (Properties of the normal distribution 2)

If $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$ are independent, then for any constants a_1, a_2,

$$a_1X_1 + a_2X_2 \sim N(a_1\mu_1 + a_2\mu_2,\ a_1^2\sigma_1^2 + a_2^2\sigma_2^2).$$

Proof. [Supplement] By independence the mgf of a sum is the product of the mgfs. Using Step 1 of the proof of Theorem 4-1,

$$\begin{aligned} M_{a_1X_1+a_2X_2}(t) &= M_{X_1}(a_1t)\, M_{X_2}(a_2t) = e^{a_1\mu_1 t + a_1^2\sigma_1^2t^2/2}\, e^{a_2\mu_2 t + a_2^2\sigma_2^2t^2/2} \\ &= \exp\left\{(a_1\mu_1 + a_2\mu_2)t + \tfrac{1}{2}(a_1^2\sigma_1^2 + a_2^2\sigma_2^2)t^2\right\}, \end{aligned}$$

the mgf of the stated normal distribution. By induction the same holds for any finite number of independent normal variables: $\sum a_iX_i \sim N(\sum a_i\mu_i, \sum a_i^2\sigma_i^2)$. ■

Example 4-5 (Calories burned)

Calories burned running each morning are approximately $N(200, 15^2)$ and calories burned cycling approximately $N(80, 5^2)$, independently. Find the probability that the total burned in a day is at least 300 kcal.
Solution. The total S is approximately normal with mean $200 + 80 = 280$ and variance $15^2 + 5^2 = 250$. So

$$P(S \geq 300) = P\Big(\frac{S-280}{\sqrt{250}} \geq \frac{300-280}{\sqrt{250}}\Big) \approx P(Z \geq 1.27) = 1 - 0.898 = 0.102.$$

4.2 Sampling distributions

Motivation. A single estimate, say a sample proportion of 76.4%, says nothing by itself about its error. The accuracy of an estimation method is judged by looking at *all* possible samples, i.e. by the probability distribution of the statistic.

4.2.1 Sampling distribution of a statistic

A survey estimates the population proportion p by the sample proportion (number in favour)/(sample size). Accuracy is expressed as the probability that the estimation error is

at most a given amount, over random selection of the sample.

Illustration: all samples of size 3

Population $\{\text{yes}, \text{no}, \text{yes}, \text{no}, \text{yes}\}$ (true $p = 0.6$), sample of size 3 without replacement.

Sample	Probability	Sample proportion	Error
{yes, yes, yes}	1/10	1	0.4
{yes, yes, no}	6/10	2/3	0.067
{yes, no, no}	3/10	1/3	−0.267

So $P(|\text{error}| \le 0.1) = 0.6$ and $P(|\text{error}| \le 0.3) = 0.9$.

Definition (Statistic and sampling distribution)

A **statistic** is a random variable computed from the sample. The probability distribution of a statistic is its **sampling distribution**; it quantifies how reliable an estimate is.

Example 4-6 (Sample mean from a small population)

From the finite population $\{2, 2, 4, 5\}$ draw a sample of size 2 without replacement. Find the distribution of the sample mean.
Solution. The six equally likely samples (probability $\frac{1}{6}$ each) and their means are

Sample	$\{2,2\}$	$\{2,4\}$	$\{2,5\}$	$\{2,4\}$	$\{2,5\}$	$\{4,5\}$
Mean	2	3	3.5	3	3.5	4.5

Hence $P(\bar{X} = 2) = \frac{1}{6}$, $P(\bar{X} = 3) = \frac{1}{3}$, $P(\bar{X} = 3.5) = \frac{1}{3}$, $P(\bar{X} = 4.5) = \frac{1}{6}$.

The sampling distribution depends on both the population distribution and the sampling method (with or without replacement).

4.2.2 Random samples

A sample drawn by simple random sampling without replacement from a finite population is a **random sample**, written $X_1, \ldots, X_n$ (X_i is the value of the ith unit; categorical answers are coded 1/0). Sampling with replacement gives $X_1, \ldots, X_n$ that

(a) each have the population distribution, and

(b) are mutually independent.

When the population is large, a random sample without replacement nearly has properties (a) and (b); this motivates the definition for infinite populations.

Definition 4-4 (Random sample)

(a) Finite population: a sample drawn by simple random sampling without replacement.
(b) Infinite population: random variables $X_1, \ldots, X_n$ such that (i) each X_i has the population distribution and (ii) $X_1, \ldots, X_n$ are independent.

Exact sampling distributions for finite populations are complicated, so a large finite population is treated as infinite. From Chapter 5 on, all inference assumes a random sample from an infinite population.

4.3 The binomial distribution

Motivation. When each unit is classified into two categories, the number of "successes" in a random sample is the basic statistic; its distribution is the binomial distribution.

Consider an infinite population of 1s and 0s with proportion p of 1s ($0 < p < 1$). A random sample $X_1, \ldots, X_n$ consists of independent Bernoulli(p) variables, i.e. n independent repeated Bernoulli trials, and the number of successes is

$$X = X_1 + \cdots + X_n.$$

Its distribution is the **binomial distribution** $B(n, p)$; $B(1, p)$ is the Bernoulli distribution.

Definition 4-5 (Binomial distribution)

With n trials and success probability p, the probability mass function is

$$p(x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad x = 0, 1, \ldots, n.$$

The **binomial coefficient** $\binom{n}{x}$ is the coefficient of $a^{n-x}b^x$ in $(a+b)^n = \sum_{x=0}^{n} \binom{n}{x} a^{n-x}b^x$ and can be read off Pascal's triangle. For $p = 0.5$ the binomial distribution is symmetric; for $p < 0.5$ it is **skewed to the right** (long right tail), for $p > 0.5$ **skewed to the left**.

Example 4-7 (Guessing on a multiple-choice test)

A test has 20 five-choice questions and every answer is guessed at random. Find the probability of
(a) no correct answer;
(b) 4 to 6 correct answers;
(c) at least 2 correct answers.
Solution. The number correct is $X \sim B(20, 0.2)$.
(a) $P(X = 0) = \binom{20}{0} 0.2^0\, 0.8^{20} = 0.012$.
(b) $P(4 \le X \le 6) = \sum_{x=4}^{6} \binom{20}{x} 0.2^x\, 0.8^{20-x} = 0.502$.
(c) $P(X \ge 2) = 1 - P(X = 0) - P(X = 1) = 1 - 0.8^{20} - 20(0.2)(0.8^{19}) = 0.931$.

Example 4-8 (On-time deliveries)

A courier delivers on time 95% of the time. In a random sample of 20 deliveries, find the probability that the sample on-time rate is at most 90%.
Solution. The number of late deliveries X is approximately $B(20, 0.05)$. An on-time rate $\le 90\%$ means a late rate $X/20 \ge 0.1$, so

$$P\Big(\frac{X}{20} \ge 0.1\Big) = P(X \ge 2) = 1 - P(X = 0) - P(X = 1) = 0.264.$$

Theorem 4-3 (Mean and variance of $B(n, p)$)

If $X \sim B(n, p)$, then
(a) $E(X) = np$;
(b) $\mathrm{Var}(X) = np(1-p)$.

Proof. Write $X = X_1 + \cdots + X_n$ with X_i independent Bernoulli(p), so $E(X_i) = p$ and $\mathrm{Var}(X_i) = p(1-p)$ (Example 4-1).

(a) By linearity of expectation, $E(X) = \sum_i E(X_i) = np$.

(b) For independent variables the covariances vanish, so $\mathrm{Var}(X) = \sum_i \mathrm{Var}(X_i) = np(1-p)$. ■

4.4 The distribution of the sample mean

Motivation. The sample mean is the basic estimator of a population mean. Its mean and variance follow from simple rules; for normal populations it is exactly normal, and by the central limit theorem it is approximately normal for any population when n is large.

For a 0/1 population the population mean is the proportion and the sample mean is the sample proportion (§4.3). Now let $X_1, \ldots, X_n$ be a random sample from an infinite population with mean μ and variance σ^2, and

$$\bar{X} = \frac{1}{n}(X_1 + \cdots + X_n).$$

Theorem 4-4 (Distribution of the sample mean 1)

For a random sample of size n from a population with mean μ and variance σ^2,
(a) $E(\bar{X}) = \mu$;
(b) $\mathrm{Var}(\bar{X}) = \sigma^2/n$;
(c) $\mathrm{sd}(\bar{X}) = \sigma/\sqrt{n}$.

Proof.

(a) Since $E(X_i) = \mu$, $E(\bar{X}) = \frac{1}{n}\{E(X_1) + \cdots + E(X_n)\} = \frac{1}{n} \cdot n\mu = \mu$.

(b) Since the X_i are independent with $\mathrm{Var}(X_i) = \sigma^2$, $\mathrm{Var}(\bar{X}) = \frac{1}{n^2}\{\mathrm{Var}(X_1) + \cdots + \mathrm{Var}(X_n)\} = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$.

(c) Take square roots in (b). ■

Finite population

For a sample of size n drawn without replacement from a finite population of size N with mean μ and variance σ^2,

$$E(\bar{X}) = \mu, \qquad \mathrm{Var}(\bar{X}) = \frac{N-n}{N-1} \cdot \frac{\sigma^2}{n}.$$

Thus $\bar{X}$ is centred at μ and concentrates around μ as n grows.

Theorem 4-5 (Distribution of the sample mean 2)

If the population is $N(\mu, \sigma^2)$, then $\bar{X} \sim N(\mu, \sigma^2/n)$.

Proof. By the extension of Theorem 4-2 to n variables, $\bar{X} = \sum_{i=1}^{n} \frac{1}{n} X_i$ is normal with mean $\sum \frac{1}{n}\mu = \mu$ and variance $\sum \frac{1}{n^2}\sigma^2 = \sigma^2/n$. ■

Theorem 4-6 (Distribution of the sample mean 3: central limit theorem)

For a random sample of size n from any population with mean μ and variance σ^2, when n is

large $\bar{X}$ is approximately $N(\mu, \sigma^2/n)$:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \dot{\sim} N(0,1).$$

Proof. [Supplement] We prove the precise statement $P(\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \le z) \to P(Z \le z)$ for every z, under the extra assumption that the mgf $M(t) = E(e^{t(X_1-\mu)/\sigma})$ is finite for $|t| < h$. We use the *continuity theorem*: if $M_{W_n}(t) \to M_W(t)$ for all $|t| < h$, then the cdf of W_n converges to that of W at every continuity point.

Step 1. Let $Y_i = (X_i - \mu)/\sigma$, so the Y_i are i.i.d. with $E(Y_i) = 0$, $E(Y_i^2) = 1$, and

$$W_n = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} Y_i, \qquad M_{W_n}(t) = \Big\{ M\Big(\frac{t}{\sqrt{n}}\Big) \Big\}^n.$$

Step 2. By Taylor's theorem, $M(s) = 1 + M'(0)s + \frac{1}{2}M''(0)s^2 + r(s)$ with $r(s)/s^2 \to 0$ as $s \to 0$. Here $M'(0) = E(Y_1) = 0$ and $M''(0) = E(Y_1^2) = 1$, so

$$M\Big(\frac{t}{\sqrt{n}}\Big) = 1 + \frac{t^2}{2n} + r\Big(\frac{t}{\sqrt{n}}\Big), \qquad n\, r\Big(\frac{t}{\sqrt{n}}\Big) = t^2 \cdot \frac{r(t/\sqrt{n})}{(t/\sqrt{n})^2} \to 0.$$

Step 3. Hence $M_{W_n}(t) = (1 + \frac{a_n}{n})^n$ with $a_n = \frac{t^2}{2} + n\, r(t/\sqrt{n}) \to \frac{t^2}{2}$, and since $(1 + a_n/n)^n \to e^a$ whenever $a_n \to a$, $M_{W_n}(t) \to e^{t^2/2}$, the mgf of $N(0,1)$.

Step 4. By the continuity theorem, and since the cdf of $N(0,1)$ is continuous, $P(W_n \le z) \to P(Z \le z)$ for every z. (Without the mgf assumption the same argument works with characteristic functions $E(e^{itY})$.) ∎

The striking point is that the conclusion holds whatever the shape of the population; this is why the normal distribution is central to inference.

Illustration 4-4: the CLT in action

The population is uniform on the digits $0, 1, \ldots, 9$ (each with probability 0.1), far from normal, with $\mu = 4.5$ and $\sigma^2 = 8.25$. One hundred random samples of size 5 were drawn and $\bar{X}$ computed for each. The histogram of the 100 sample means is roughly bell-shaped and centred near 4.5 (relative frequencies 0.29 and 0.27 in the classes $[4,5)$ and $[5,6)$, tapering to 0.03 and 0.07 in the outer classes), even though $n = 5$. Larger n brings it still closer to normal.

How large n must be depends on how far the population is from normal. For a normal population $\bar{X}$ is exactly normal; otherwise $n > 30$ usually gives a satisfactory normal approximation.

Example 4-9 (Calories of a new drink)

A new drink has mean 168 kcal and standard deviation 6 kcal. For a random sample of 100 drinks, find $P(167 \le \bar{X} \le 169)$.
Solution. With $n = 100$ the CLT applies; $\bar{X}$ has mean 168 and standard deviation $6/\sqrt{100} = 0.6$. So

$$\begin{aligned} P(167 \le \bar{X} \le 169) &\approx P\Big(\frac{167 - 168}{0.6} \le Z \le \frac{169 - 168}{0.6}\Big) \\ &= P(-1.67 \le Z \le 1.67) = 0.9525 - 0.0475 = 0.9050. \end{aligned}$$

4.4.1 Normal approximation to the binomial

If $X_1, \ldots, X_n$ is a random sample from $B(1,p)$, then $X = X_1 + \cdots + X_n \sim B(n,p)$, the population mean is p, the variance is $p(1-p)$ and the sample mean is the sample proportion $\hat{p} = X/n$. The CLT therefore gives the following.

Definition (Normal approximation to the binomial)

If $X \sim B(n,p)$ and n is large (rule of thumb: $np \geq 5$ and $n(1-p) \geq 5$), then X is approximately $N(np, np(1-p))$:

$$\frac{X - np}{\sqrt{np(1-p)}} = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \dot{\sim} N(0,1).$$

For example $B(15, 0.6)$ is already well approximated by $N(9, 3.6)$ since $np = 9 > 5$.

Example 4-10 (Normal approximation)

Let $X \sim B(100, \frac{1}{5})$. Approximate
(a) $P(X \leq 21)$;
(b) $P(15 \leq X \leq 19)$.
Solution. $np = 20$ and $\sqrt{np(1-p)} = \sqrt{100 \cdot \frac{1}{5} \cdot \frac{4}{5}} = 4$.
(a) $P(X \leq 21) \approx P(Z \leq \frac{21-20}{4}) = P(Z \leq 0.25) = 0.5987$.
(b) $P(15 \leq X \leq 19) \approx P(-1.25 \leq Z \leq -0.25) = 0.4013 - 0.1056 = 0.2957$.

Continuity correction

A binomial probability is a sum of bars of width 1 centred at the integers. For $X \sim B(15, 0.6)$, $P(7 \leq X \leq 10)$ is better approximated by the normal area from 6.5 to 10.5:

$$\begin{aligned} P(7 \leq X \leq 10) = P(6.5 \leq X \leq 10.5) &\approx P\Big(\frac{6.5-9}{\sqrt{3.6}} \leq Z \leq \frac{10.5-9}{\sqrt{3.6}}\Big) \\ &= P(-1.318 \leq Z \leq 0.791) = 0.692, \end{aligned}$$

very close to the exact value 0.688. This adjustment is the **continuity correction**.

4.5 The chi-square, t and F distributions

Motivation. Inference about a normal variance, and about a normal mean when σ is unknown, relies on statistics built from the sample variance. Their sampling distributions are the chi-square, t and F distributions.

For a random sample $X_1, \ldots, X_n$ from $N(\mu, \sigma^2)$ the **sample variance** is

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2.$$

4.5.1 The chi-square distribution

Definition 4-6 (Chi-square distribution)

If $Z_1, \ldots, Z_k$ is a random sample from $N(0,1)$, the distribution of $Z_1^2 + \cdots + Z_k^2$ is the **chi-square distribution with k degrees of freedom**:

$$Z_1^2 + \cdots + Z_k^2 \sim \chi^2(k).$$

Because each Z_i^2 is mostly near 0, the chi-square density is concentrated on the left with a long right tail. The upper quantile $\chi^2_\alpha(k)$ is defined by $P(V \geq \chi^2_\alpha(k)) = \alpha$ for $V \sim \chi^2(k)$; e.g. $\chi^2_{0.995}(4) = 0.21$, $\chi^2_{0.975}(4) = 0.48$, $\chi^2_{0.05}(4) = 9.49$.

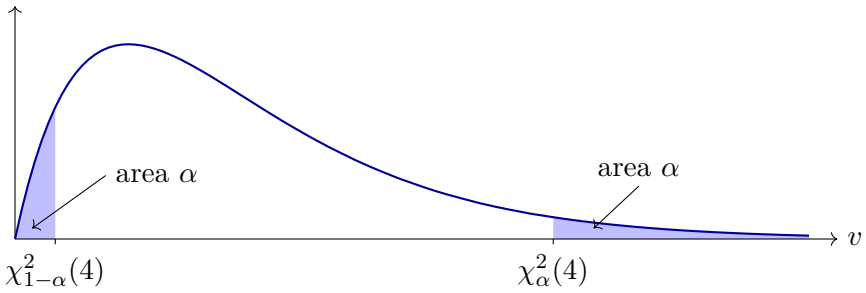


Figure 2: The $\chi^2(4)$ density $ve^{-v/2}/4$ with $\alpha = 0.05$: $\chi^2_{0.95}(4) = 0.71$, $\chi^2_{0.05}(4) = 9.49$.

Theorem 4-7 (Additivity of the chi-square distribution)

If $V_1 \sim \chi^2(k_1)$ and $V_2 \sim \chi^2(k_2)$ are independent, then $V_1 + V_2 \sim \chi^2(k_1 + k_2)$.

Proof. [Supplement]

Step 1. For $Z \sim N(0,1)$ and $t < \frac{1}{2}$, substituting $u = z\sqrt{1-2t}$,

$$E(e^{tZ^2}) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-(1-2t)z^2/2}\, dz = (1-2t)^{-1/2}.$$

Step 2. Hence, by independence, $V \sim \chi^2(k)$ has mgf $M_V(t) = (1-2t)^{-k/2}$, $t < \frac{1}{2}$; conversely any variable with this mgf has the $\chi^2(k)$ distribution (uniqueness of mgfs).

Step 3. For independent V_1, V_2, $M_{V_1+V_2}(t) = (1-2t)^{-k_1/2}(1-2t)^{-k_2/2} = (1-2t)^{-(k_1+k_2)/2}$, so $V_1 + V_2 \sim \chi^2(k_1 + k_2)$. ■

Theorem 4-8 (Distribution of the sample variance from a normal population)

For a random sample $X_1, \ldots, X_n$ from $N(\mu, \sigma^2)$ with sample variance S^2,

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1).$$

Moreover $\bar{X}$ and S^2 are independent.

Proof. The book's argument: since $(X_i - \mu)/\sigma \sim N(0,1)$ independently, $\sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma}\right)^2 \sim \chi^2(n)$. Adding and subtracting $\bar{X}$ (the cross term vanishes because $\sum(X_i - \bar{X}) = 0$),

$$\sum_{i=1}^{n} \left(\frac{X_i - \mu}{\sigma}\right)^2 = \frac{(n-1)S^2}{\sigma^2} + \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}\right)^2,$$

and the last term is $\chi^2(1)$. "Subtracting" degrees of freedom gives $\chi^2(n-1)$. We make this rigorous.

[Supplement]

Step 1. Let $Z_i = (X_i - \mu)/\sigma$, independent $N(0,1)$, so $\mathbf{Z} = (Z_1, \dots, Z_n)^\top$ has joint density $(2\pi)^{-n/2} e^{-\|\mathbf{z}\|^2/2}$. Let A be an $n \times n$ orthogonal matrix whose first row is $(1/\sqrt{n}, \dots, 1/\sqrt{n})$ (complete this unit vector to an orthonormal basis), and $\mathbf{W} = A\mathbf{Z}$.

Step 2. The map $\mathbf{z} \mapsto A\mathbf{z}$ has Jacobian $|\det A| = 1$ and preserves $\|\mathbf{z}\|$, so $\mathbf{W}$ has the same density: $W_1, \dots, W_n$ are independent $N(0,1)$.

Step 3. $W_1 = \sqrt{n}\,\bar{Z}$ and $\sum_{i=1}^n W_i^2 = \sum_{i=1}^n Z_i^2$, hence

$$\sum_{i=2}^{n} W_i^2 = \sum_{i=1}^{n} Z_i^2 - n\bar{Z}^2 = \sum_{i=1}^{n} (Z_i - \bar{Z})^2 = \frac{(n-1)S^2}{\sigma^2}.$$

Step 4. By Definition 4-6 this is $\chi^2(n-1)$. It is a function of $W_2, \dots, W_n$, while $\bar{X} = \mu + \sigma W_1/\sqrt{n}$ is a function of W_1; so $\bar{X}$ and S^2 are independent. ■

Next take independent random samples $X_1, \dots, X_{n_1}$ from $N(\mu_1, \sigma^2)$ and $Y_1, \dots, Y_{n_2}$ from $N(\mu_2, \sigma^2)$ (common variance), with

$$S_1^2 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (X_i - \bar{X})^2, \qquad S_2^2 = \frac{1}{n_2 - 1} \sum_{i=1}^{n_2} (Y_i - \bar{Y})^2.$$

Definition (Pooled sample variance)

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2},$$

the weighted average of S_1^2, S_2^2 with weights $\frac{n_j - 1}{(n_1 - 1) + (n_2 - 1)}$; it estimates the common variance σ^2.

Theorem 4-9 (Sample variances from two normal populations with equal variance)

Under the above assumptions,

$$\frac{(n_1 + n_2 - 2)S_p^2}{\sigma^2} \sim \chi^2(n_1 + n_2 - 2).$$

Proof. By Theorem 4-8, $(n_1 - 1)S_1^2/\sigma^2 \sim \chi^2(n_1 - 1)$ and $(n_2 - 1)S_2^2/\sigma^2 \sim \chi^2(n_2 - 1)$, and they are independent since they come from independent samples. Their sum is $(n_1 + n_2 - 2)S_p^2/\sigma^2$, which is $\chi^2(n_1 + n_2 - 2)$ by Theorem 4-7. ■

4.5.2 The t distribution

When σ is unknown we replace it by the sample standard deviation S and obtain the **studentized** sample mean $\frac{\bar{X} - \mu}{S/\sqrt{n}}$.

Definition 4-7 (t distribution)

If $Z \sim N(0,1)$ and $V \sim \chi^2(k)$ are independent, the distribution of

$$T = \frac{Z}{\sqrt{V/k}}$$

is the t **distribution with k degrees of freedom**, $T \sim t(k)$.

Like $N(0,1)$, $t(k)$ is symmetric about 0, but it has heavier tails; it is used especially for small samples. The upper quantile $t_\alpha(k)$ satisfies $P(T \geq t_\alpha(k)) = \alpha$; e.g. $t_{0.1}(4) = 1.533$, $t_{0.05}(4) = 2.132$, $t_{0.01}(4) = 3.747$.

Theorem 4-10 (Distribution of the studentized sample mean)

For a random sample $X_1, \ldots, X_n$ from $N(\mu, \sigma^2)$,

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1).$$

Proof. Dividing numerator and denominator by $\sigma/\sqrt{n}$,

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} = \frac{(\bar{X} - \mu)/(\sigma/\sqrt{n})}{\sqrt{\dfrac{(n-1)S^2}{\sigma^2} \Big/ (n-1)}}.$$

The numerator is $N(0,1)$ (Theorem 4-5), the quantity under the root is $\chi^2(n-1)$ divided by $n-1$ (Theorem 4-8), and they are independent because $\bar{X}$ and S^2 are (Theorem 4-8; the book omits this proof). By Definition 4-7 the ratio is $t(n-1)$. ■

For two normal populations with equal variance, a short computation gives

$$\frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{\dfrac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sigma\sqrt{1/n_1 + 1/n_2}}}{\sqrt{\dfrac{(n_1 + n_2 - 2)S_p^2}{\sigma^2} \Big/ (n_1 + n_2 - 2)}}.$$

Theorem 4-11 (t distribution for two normal populations with equal variance)

Under the assumptions of Theorem 4-9,

$$\frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t(n_1 + n_2 - 2).$$

Proof.

Step 1. Numerator. $\bar{X} \sim N(\mu_1, \sigma^2/n_1)$ and $\bar{Y} \sim N(\mu_2, \sigma^2/n_2)$ are independent, so by Theorem 4-2 $\bar{X} - \bar{Y} \sim N(\mu_1 - \mu_2, \sigma^2(\frac{1}{n_1} + \frac{1}{n_2}))$, and the standardized numerator in the display is $N(0,1)$.

Step 2. Denominator. By Theorem 4-9, $(n_1 + n_2 - 2)S_p^2/\sigma^2 \sim \chi^2(n_1 + n_2 - 2)$.

Step 3. Independence. By Theorem 4-8, $\bar{X}$ is independent of S_1^2 and $\bar{Y}$ of S_2^2; as the two samples are independent, $(\bar{X}, \bar{Y})$ is independent of (S_1^2, S_2^2), hence of S_p^2.

Step 4. Definition 4-7 gives $t(n_1 + n_2 - 2)$. ■

4.5.3 The F distribution

Definition 4-8 (F distribution)

If $V_1 \sim \chi^2(k_1)$ and $V_2 \sim \chi^2(k_2)$ are independent, the distribution of

$$F = \frac{V_1/k_1}{V_2/k_2}$$

is the F **distribution with** (k_1, k_2) **degrees of freedom**, $F \sim F(k_1, k_2)$.

The F density is right-skewed on $(0, \infty)$. Its upper quantile $F_\alpha(k_1, k_2)$ satisfies $P(F \geq F_\alpha(k_1, k_2)) = \alpha$; e.g. $F_{0.10}(3, 4) = 4.19$, $F_{0.05}(3, 4) = 6.59$.

Theorem 4-12 (Property of the F distribution)

If $F \sim F(k_1, k_2)$, then $1/F \sim F(k_2, k_1)$. Consequently

$$F_{1-\alpha}(k_2, k_1) = \frac{1}{F_\alpha(k_1, k_2)}.$$

Proof.

Step 1. Write $F = \frac{V_1/k_1}{V_2/k_2}$ as in Definition 4-8. Then $1/F = \frac{V_2/k_2}{V_1/k_1}$ is a ratio of the same form with the roles of (V_1, k_1) and (V_2, k_2) exchanged, so $1/F \sim F(k_2, k_1)$.

Step 2. Let $F' = 1/F \sim F(k_2, k_1)$. Since $F > 0$, $\alpha = P(F \geq F_\alpha(k_1, k_2)) = P(F' \leq 1/F_\alpha(k_1, k_2))$, so $P(F' \geq 1/F_\alpha(k_1, k_2)) = 1-\alpha$ (F' is continuous). By definition of the upper quantile, $F_{1-\alpha}(k_2, k_1) = 1/F_\alpha(k_1, k_2)$. ■

For example, $F_{0.95}(4, 3) = 1/F_{0.05}(3, 4) = 1/6.59 = 0.15$.

For independent samples from $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$, $(n_1 - 1)S_1^2/\sigma_1^2 \sim \chi^2(n_1 - 1)$ and $(n_2 - 1)S_2^2/\sigma_2^2 \sim \chi^2(n_2 - 1)$ independently.

Theorem 4-13 (Ratio of sample variances from two normal populations)

Under these assumptions,

$$\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n_1 - 1, n_2 - 1).$$

Proof. Put $V_j = (n_j - 1)S_j^2/\sigma_j^2 \sim \chi^2(n_j - 1)$ (Theorem 4-8); V_1, V_2 are independent. Then $\frac{V_1/(n_1-1)}{V_2/(n_2-1)} = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}$, which is $F(n_1 - 1, n_2 - 1)$ by Definition 4-8. ■

Theorem 4-14 (Relation between the F and t distributions)

If $T \sim t(k)$, then $T^2 \sim F(1, k)$.

Proof. Write $T = Z/\sqrt{V/k}$ with $Z \sim N(0, 1)$, $V \sim \chi^2(k)$ independent. Then $T^2 = \frac{Z^2/1}{V/k}$, where $Z^2 \sim \chi^2(1)$ (Definition 4-6) and Z^2 is independent of V. By Definition 4-8, $T^2 \sim F(1, k)$. ■

4.6 Normal quantile plots

Motivation. Most inference in later chapters assumes a normal population. A normal quantile (Q–Q) plot is a quick graphical check of that assumption.

A q-quantile of a distribution divides its total area in the ratio $q : (1-q)$. A **normal quantile plot** (Q–Q plot) plots the quantiles of the standard normal distribution against the corresponding quantiles of the data.

Definition (Normal quantile plot)

Order the data $x_{(1)} < x_{(2)} < \cdots < x_{(n)}$. The value $x_{(i)}$ is roughly the $\frac{i}{n+1}$-quantile of the data, whose standard normal counterpart is $z_{1-i/(n+1)}$. Plot the points

$$\big(z_{1-i/(n+1)},\, x_{(i)}\big), \qquad i = 1, \ldots, n.$$

If the points lie close to a straight line, the normality assumption is reasonable. Software often uses $\frac{i-3/8}{n+1/4}$ instead of $\frac{i}{n+1}$.

For instance, with $n = 99$ observations from $N(2, 3^2)$, the population quantiles are $3z_{0.99} + 2$, $3z_{0.98} + 2, \ldots, 3z_{0.01} + 2$, a linear function of the standard normal quantiles, so the points $(z_{0.99}, x_{(1)}), \ldots, (z_{0.01}, x_{(99)})$ lie near a line with slope 3 and intercept 2.

Reading a Q–Q plot

Systematic departures from the line indicate a non-normal population:
(a) light tails: the extreme points bend toward the centre (an S-shape flatter than the line at both ends);
(b) heavy tails: the extreme points fall below the line on the left and above it on the right;
(c) long right tail: the points curve upward on the right;
(d) long left tail: the points curve downward on the left.